

Ying Xu

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OBJECTIVE

PhD candidate in Power Systems / Wind Energy (Fall 2027) — seeking a fully funded position in renewable-energy integration, power-system simulation, or grid interfacing.

EDUCATION

North China Electric Power University 2023.09 – 2027.06 (expected)

B.Eng. in New Energy Science and Engineering (wind-energy focus)

- GPA 82.9% (3.29/5.00). Top courses: Turbine Aerodynamics 95, Wind Resource Measurement & Assessment 95, Numerical Simulation 90, Reliability Engineering 90; also Circuit Analysis, Electrical Machines, Power Electronics, Control Theory

AWARDS & HONORS

- **National First Prize**, China Undergraduate Electronic Design Contest (2025); nominated for the TI Cup
- National Third Prize, China Undergraduate Electronic Design Contest — TI Cup Analog Electronic System Design (2024)
- University Merit Student; 2nd-Class University Scholarship; Sci-Tech Innovation Scholarship

RESEARCH EXPERIENCE

Direct-Drive PMSG Wind-Turbine Multi-Timescale Modeling & Simulation (in progress) 2026.10 – 2027.03

National Key Laboratory of Renewable Energy Electric Systems, NCEPU — funded undergraduate research project, sole researcher (advised by Dr. Zhao)

- Building a hierarchical multi-timescale model of a direct-drive PMSG wind turbine — rotor aerodynamics and drivetrain (seconds scale), DC-link voltage / power / MPPT control (hundreds of ms), current control and electromagnetic dynamics (ms) — for system-level grid-integration simulation on the lab's in-house power-system simulation platform
- Deriving reduced-order average-value models via sensitivity analysis and singular-perturbation order reduction; benchmarking the model structure against WECC Type 4 and IEC 61400-27-1 (Type 4A/4B) reference models
- Completed so far: systematic literature review (12 papers spanning Type 4/4A/4B standards, model-order reduction, LVRT and primary-frequency-response control) and the overall modeling scheme; ongoing: model derivation, parameter tables, platform implementation, and wind-speed-step / voltage-sag / frequency-disturbance case studies

CFD Simulation Researcher 2025.07 – 2025.08

Wind Power Research Center, School of New Energy — independent project under a PhD researcher's guidance

- Ran five external-flow cases in ANSYS Fluent (cylinder flow, vortex-induced vibration, NACA0012 static/dynamic stall, S809 dynamic stall) with an OpenFOAM workflow; built meshes, set turbulence models and inflation layers, post-processed airfoil force coefficients and stall behavior

SELECTED PROJECT

Monocular-Vision Target Measurement Device — National First Prize 2025.07 – 2025.08

China Undergraduate Electronic Design Contest, National Finals (Hardware & Integration)

- Single 4K industrial camera measures target dimensions and distance (rectangle/circle/triangle), error < 2 mm at 1.6 m
- Designed a custom power-monitoring circuit (commercial modules prohibited) tracking real-time and peak consumption: peak 7 W, idle 5 W (limit 10 W); owned MIPI display-module adaptation (custom adapter board) and system integration, supported the vision-pipeline and UI work
- Platform: RK3576 embedded Linux + Flutter front end / Python-OpenCV back end (separated architecture, later ported to smartphone)

PERSONAL PROJECTS

12 V Power-Distribution Monitor 2026.09 – present

- End-to-end power-monitoring system: custom open-source PCB, ESP32-C3 firmware (C/ESP-IDF) sampling an INA226 at 10 Hz over a custom 20-byte TCP wire protocol, with adaptive 10 Hz / 0.1 Hz sampling by viewer presence
- **Dual-path overcurrent protection** (hardware ALERT-pin comparator + independent backend threshold) with per-event time/voltage/current/power logging; **dual-slot OTA with automatic rollback** and a custom two-slot partition table (a bad update cannot brick a deployed device)
- Python backend (FastAPI, async TCP ingest, SQLite persistence, WebSocket live broadcast) + web dashboard; INA226 shunt calibration sets all current/power readings

Wind-Resource Assessment Toolkit 2026.06

- Python pipeline for wind-farm siting: multi-stage automated QC on SODAR/LIDAR radar (40–200 m) and met-mast (2–80 m) wind data (range / correlation / trend / missing-value checks)
- Weibull distribution fitting, power-law wind-shear and air-density estimation, wind-energy-density wind roses, and per-height data-availability statistics; unit-tested

NanoScope — Two-Channel Pocket Oscilloscope 2025.06 – 2025.07

- Custom measurement instrument: OPA4197 analog front end (–5 V to +12 V), STM32 ADC+DMA high-speed acquisition, and frequency/**phase** measurement via external timers (50 Hz–200 kHz); ESP32 real-time display + BLE Flutter app

LEADERSHIP

Head of Technology Department, MOS Innovation Lab, NCEPU 2025.05 – present

- Organized STM32 embedded-development training and guided new members through a line-following robot; designed the lab's technical-assessment scheme and led project reviews/defenses for dozens of members

TECHNICAL SKILLS

Simulation	ANSYS Fluent (2D external flow), OpenFOAM, 3D modeling
Programming	C, Python (embedded & backend), MATLAB (basic)
Embedded	STM32 / ESP32 (ADC, UART, SPI, I ² C, PWM, DMA), Rockchip Linux, MIPI display, PCB design
Tools	Git, Docker, Linux, AutoCAD, SolidWorks
Languages	English: IELTS 7.0 (fluent)